Polynomial-calculus degree lower bounds for the rank and inversion principles

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Editorial note. References to unselected original sections are printed with their accurate original article chapter numbers; selected section headers and original labels remain in place. No absent section is presented as an included proof.

Allowed standard prerequisites (editorial).

- Known graph-isomorphism and functional-pigeonhole polynomial-calculus lower bounds; effective Nullstellensatz and tensor-rank inheritance. Exact original cited statements retained where used; standard earlier literature not recursively reproved

Abstract

The Tensor Isomorphism problem (TI) has recently emerged as having connections to multiple areas of research within complexity and beyond, but the current best upper bound is essentially the brute force algorithm. Being an algebraic problem, TI (or rather, proving that two tensors are *non*-isomorphic) lends itself very naturally to algebraic and semi-algebraic proof systems, such as the Polynomial Calculus (PC) and Sum of Squares (SoS). For its combinatorial cousin Graph Isomorphism, essentially optimal lower bounds are known for approaches based on PC and SoS (Berkholz & Grohe, SODA '17). Our main results are an $\Omega(n)$ lower bound on PC degree or SoS degree for Tensor Isomorphism, and a nontrivial upper bound for testing isomorphism of tensors of bounded rank.

We also show that PC cannot perform basic linear algebra in sublinear degree, such as comparing the rank of two matrices, or deriving $BA = I$ from $AB = I$. As linear algebra is a key tool for understanding tensors, we introduce a strictly stronger proof system, PC+Inv, which allows as derivation rules all substitution instances of the implication $AB = I \to BA = I$. We conjecture that even PC+Inv cannot solve TI in polynomial time either, but leave open getting lower bounds on PC+Inv for any system of equations, let alone those for TI. We also highlight many other open questions about proof complexity approaches to TI.

1 Introduction

Tensors have rapidly emerged as a fundamental data structure and key mathematical object of the 21st century. They play key roles in many different areas of science, engineering, and mathematics, from quantum mechanics and general relativity to neural networks [NPOV15] and mechanical engineering. They arise in theoretical computer science in many ways, including from (post-quantum) cryptography [Pat96, JQSY19], derandomization, Matrix Multiplication, Graph Isomorphism [GQ21b], and several different parts of Geometric Complexity Theory.

The fundamental notion of equivalence between tensors is that of isomorphism: two tensors are isomorphic if one can be transformed into the other by an invertible linear change of basis in each of the corresponding vector spaces. For example, two 2-tensors (=matrices) M, M' are equivalent under this notion if there are invertible matrices X, Y such that $XMY = M'$; similarly, two 3-tensors, represented by 3-way arrays T_{ijk}, T'_{ijk} are isomorphic if there are three invertible matrices X, Y, Z such that

$$\sum_{ijk} X_{ii'} Y_{jj'} Z_{kk'} T_{ijk} = T'_{i'j'k'} \tag{1}$$

for all i', j', k'. The problem of (3-)Tensor Isomorphism (TI) is: given two such 3-way arrays, to decide if they are isomorphic.

Over finite fields, two different versions of TI sandwich the complexity of its more famous cousin, Graph Isomorphism. Namely, as presented above, GI reduces to TI. In the other direction, over a finite field $\mathbb{F} = \mathbb{F}_{p^a}$, one can take an $n \times n \times n$ tensor and list it out "verbosely", as a set of p^{an} many $n \times n$ matrices over $\mathbb{F}$; the isomorphism problem for such verbosely given tensors is equivalent to Group Isomorphism for a certain class of p-groups, widely believed to be the hardest cases of Group Isomorphism in general. As such, this verbose version of TI reduces to GI. Furthermore, with Babai's quasi-polynomial-time algorithm [Bab16], the running times are quite close: $N^{O(\log N)}$ for VerboseTI and $N^{O(\log^2 N)}$ for GI (the exponent of the exponent was worked out by Helfgott [Hel19]). Thus TI stands as a key obstacle to putting GI into P.

In this paper, we initiate the study of (algebraic) proof complexity approaches to proving that two tensors are non-isomorphic. Lower bounds on the Polynomial Calculus proof system imply lower bounds on Gröbner basis techniques, and the latter are some of the leading methods for solving TI-complete problems in cryptanalysis, e.g., [TDJ+22, FP06]. In the context of GI, proof complexity plays an important role, through its connection with the Weisfeiler–Leman (WL) algorithm. Although this algorithm does not, on its own, solve GI in polynomial time [CFI92], it is a key subroutine in many of the best algorithms for GI, both in theory [Bab16] and in practice (see [McK81, MP14]). And the picture that has emerged is that some proof systems for GI are known to be equivalent in power to WL [AM13, BG15], and some lower bounds on proof systems are closely related to lower bounds for WL [SSC14, OWWZ14, BG15]. Versions of WL for groups, and in particular finite p-groups—and hence, by the connection above, tensors over finite fields—have only recently begun to be explored [BGL+19, BS20, BS22, CL22].

1.1 Main results

We focus on the Polynomial Calculus (PC, or Gröbner) proof system [CEI96], though our results will also hold for semi-algebraic proof systems such as Sum-of-Squares [Las01] as well. PC is used to show that a system of polynomial equations over a field $\mathbb{F}$ is unsatisfiable over the algebraic closure $\overline{\mathbb{F}}$, by deriving from the system of equations, in a line-by-line fashion, the contradiction $1 = 0$. The degree of a PC proof is the maximum degree of any line appearing in the proof, and it is a fundamental result that PC proofs of constant degree can be found in polynomial time [CEI96]. Much as WL informally "captures all combinatorial approaches" to GI, PC informally "captures all approaches based on Gröbner bases" to showing that a system of polynomial equations is unsatisfiable.

The systems of equations we study are, for two non-isomorphic tensors T, T', the equations (1) along with new matrices X', Y', Z', and equations saying that these are the inverses of X, Y, Z, resp., viz.: $XX' = X'X = \mathrm{Id}$, and similarly for the others. The reason for introducing these new matrices, despite their not appearing in (1), is that these invertibility equations are only degree 2. In contrast, if we instead used the determinant to indicate that X was invertible, then our starting equations would have degree $n+1$, rather than constant degree ≤ 3. Since the main complexity measure we study on PC is degree, having starting equations of degree n would make it difficult to make meaningful lower bound statements.

Our first main result is (two proofs of) a lower bound on such techniques.

Theorem 1.1. *Over any field, there are instances of $n \times n \times n$ Tensor Isomorphism that require PC degree $\Omega(n)$ to refute. Over $\mathbb{R}$, they also require Sum-of-Squares degree $\Omega(n)$ to refute.*

The preceding goes by reduction from known lower bounds on PC for Graph Isomorphism [BG15, BG17], but has the disadvantage (from the tensor point of view) that the resulting tensors are quite sparse: in one direction, one of the slices is supported on an $\Omega(n) \times n$ matrix and all the others slices have support size 1. In a second proof (Section 6 (in the original article)), we get a polynomially worse lower bound $\Omega(\sqrt[4]{n})$, but with a reduction from Random 3XOR that is more direct. Indeed, we show that 3XOR itself can be viewed as a particular instance of a tensor problem *without* gadgets; gadgets are only then needed to reduce from that tensor problem to Tensor Isomorphism itself. In contrast, the lower bounds on PC for GI (*ibid.*) already use the Cai–Fürer–Immerman gadgets [CFI92] to reduce from XOR-SAT, and then even further gadgets are needed to reduce from GI to TI.

Our technical contributions in the above theorem are thus three-fold:

1. We show that the known reductions from GI to TI can be carried out in low-degree PC;
2. We realize 3XOR very naturally as a tensor problem; and
3. We give new reductions from 3XOR, through a series of tensor-related problems, to TI, that work as many-one reductions of the decision problems that can be carried out in low-degree PC.

Complementing our lower bound, we also show that tensors of low rank are comparatively easy to test for (non)-isomorphism. Here, one of our upper bounds is in the weaker Nullstellensatz proof system (giving a stronger upper bound than only a PC upper bound). In the Nullstellensatz proof system, a proof that a system of equations $f_1 = \cdots = f_m = 0$ is unsatisfiable consists of polynomials g_i such that $\sum g_i f_i = 1$, and the Nullstellensatz degree is the maximum degree of any $g_i f_i$. The PC degree is always at most the Nullstellensatz degree, and the gap between the two can be nearly maximal for Boolean equations: $O(1)$ versus $\Omega(n/\log n)$ [BCIP02]. (For Boolean equations, there is always an $O(n)$ upper bound, though this does not apply to TI, see Remark 1.3 below).

Theorem 1.2. *Over any field, the Nullstellensatz degree of refuting isomorphism of two $n \times n \times n$ tensors of tensor rank $\leq r$ is at most $2^{O(r^2)}$. If working over a finite field $\mathbb{F}_q$ and including the equations $x^q - x$, the PC degree is at most $O(qr^2)$.*

In particular, isomorphism of constant-rank tensors can be decided in polynomial time.

Remark 1.3. In many settings in proof complexity, Boolean axioms such as $x_i^2 = x_i$ or $x_i^2 = 1$ are included among the system of equations, and all such unsatisfiable systems of equations can be refuted in degree $O(n)$ ($n =$# variables). If this were the case here, the above would only be interesting for very small values of r. In contrast, the equations for TI do not include any such Boolean axioms, and as such the naive degree upper bound is exponential in the number of variables. For $n \times n \times n$ tensors, this gives an upper bound of $2^{O(n^2)}$ [Som99], and thus, Theorem 1.2 gives nontrivial upper bounds all the way up to $r \leq n$. (We note that $n \times n \times n$ tensors can have rank up to $\Theta(n^2)$ [Lic85].) The proof of Theorem 1.2 shows that for rank-r tensors, TI can essentially be reduced to a system of equations in only $O(r^2)$ variables.

Remark 1.4. For fixed r, testing if an $n \times n \times n$ tensor has rank $\leq r$ can be done in polynomial time, as follows. This will show that the algorithm of Theorem 1.2 genuinely solves the decision problem, and not just a promise problem. Given an $n \times n \times n$ tensor T, consider its three $n \times n^2$ flattenings. Use Gaussian elimination to put each such flattening, separately, into reduced row echelon form. If any of these flattenings has rank $> r$, reject. Otherwise, we get from this a list of $3r$ vectors $u_1, \ldots, u_r, v_1, \ldots, v_r, w_1, \ldots, w_r$, such that T lives in the $r \times r \times r$-dimensional space $\mathrm{Span}\{u_1, \ldots, u_r\} \otimes \mathrm{Span}\{v_1, \ldots, v_r\} \otimes \mathrm{Span}\{w_1, \ldots, w_r\}$. Now in this space we can write down the Brent equations [Bre70] for T to have rank $\leq r$, which will be r^3 cubic equations in $3r^2$ variables (Brent's equations [Bre70, (5.06)] were specifically for the matrix multiplication tensor, but analogous equations are easily constructed for arbitrary tensors using the same idea). Since r is constant, these equations may be solved in polynomial time (here we assume that we are either working over a finite field, a finite-degree extension of the rationals—see, for example, Grigoriev [Gri13]—or in the BSS model over an arbitrary field).

Lastly, one may wonder why we focus on 3-Tensor Isomorphism, and not some of its many related variants. Indeed, just as there are other equivalence notions for matrices—such as conjugacy XMX^{-1} and congruence XMX^T—there are many different kinds of multilinear objects that can be represented by

multi-way arrays, including tensors, homogeneous polynomials (commutative or noncommutative), alternating matrix spaces, multilinear maps, and so on, each with their own corresponding notion of isomorphism. While these problems are indeed distinct, they are all equivalent under polynomial-time isomorphisms [FGS19, GQ21b]; such problems are called TI-complete. Even isomorphism of k-way tensors (for any fixed $k \geq 3$) is equivalent to isomorphism of 3-tensors [GQ21b]. This partially justifies our focus on 3-Tensor Isomorphism. In the course of proving our reductions for the results stated above, we use many of the gadgets from [FGS19, GQ21b], and show that such uses also often yield proof complexity reductions as well. Because of the variety of gadgets used in our reductions, we believe that many, if not all, of the gadgets from those results would also yield proof complexity reductions, so the proof complexity of all the known TI-complete problems should be polynomially related.

1.2 Comparison with linear algebra, a new proof system, and a conjecture

As linear algebra is part of the core toolkit for understanding tensors, it is natural to wonder how linear algebra can help in algebraic proof complexity approaches to TI. We believe that even if it had the "full power" of linear algebra at its disposal "for free," PC could still not solve TI efficiently. We begin to make this precise in this section.

Some basic derivations in linear algebra are to relate the ranks of two matrices and to derive $BA = I$ from $AB = I$ (the Inversion Principle, one of the so-called "hard matrix identities" [SC04], only recently shown to have short NC^2-Frege proofs [HT15]). Soltys [Sol01] and Soltys & Cook [SC04] discuss the relationship between these and other standard implications in linear algebra. We show that PC is not strong enough to prove these in low-degree:

Theorem 1.5. *The unsatisfiable system of equations $XY = \mathrm{Id}_n$ where X is $n \times r$ and Y is $r \times n$ with $1 \leq r < n$, requires degree $\geq r/2 + 1$ to refute in PC, over any field.*

We refer to this system of equations as the Rank Principle, as refuting them amounts to showing that $\mathrm{rk}\,\mathrm{Id}_n > r$.

Theorem 1.6. *Any PC derivation of $BA = I$ from $AB = I$, where A, B are $n \times n$ matrices with $\{0, 1\}$ entries, requires degree $\geq n/2 + 1$, over any field.*

We also observe that the Rank Principle can be derived in low degree from the Inversion Principle.

Although it remains open whether the Inversion Principle is "complete" for linear-algebraic reasoning (see [Sol01, SC04]), we introduce the proof system PC+Inv in an attempt to capture some linear-algebraic reasoning that seems potentially useful for TI. PC+Inv has all the same derivation rules as PC, but in addition, for any square matrices A, B (whose entries may themselves be polynomials—that is, we allow substitution instances), we have the rule

$$\frac{AB = I}{BA = I}.$$

where the antecedent represents the set of n^2 equations corresponding to $AB = I$, and similarly the consequent denotes the set of n^2 equations $BA = I$ (see

2.3 for more details). Degree is still measured in the usual way, but this rule lets us "cut out" the high-degree proof that would usually be required to derive $BA = I$ from $AB = I$. We now formalize our intuition that linear algebra should not suffice to solve TI efficiently in the following:

Conjecture 1.7. TENSOR ISOMORPHISM for $n \times n \times n$ tensors requires degree $\Omega(n)$ in PC+Inv, over any field.

Despite the conjecture, we do not yet know how to prove lower bounds on PC+Inv for any unsatisfiable system of equations, let alone those coming from TI. Mod p counting principles (for p different from the characteristic of the field) strike us as potentially interesting instances to examine for PC+Inv lower bounds, before tackling a harder problem like TI. In the final section, we highlight many other open questions around the proof complexity of TI.

1.3 Organization

In Section 2 we cover preliminaries. In Section 3 we prove the lower bounds on linear algebraic principles just discussed. In Section 4 (in the original article) we prove the upper bound for isomorphism of bounded rank tensors (Theorem 1.2). In Section 5 (in the original article) we prove Theorem 1.1 by reduction from GI. In Section 6 (in the original article) we prove the polynomially related lower bound by direct reduction from RANDOM 3XOR.

2 Preliminaries

2.1 Proof systems

All our rings are commutative and unital. *Polynomial calculus* (PC) is a proof system to prove that a given system of (multivariate) polynomial equations $\mathcal{P}$ over a field $\mathbb{F}$ of the form $p = 0$, has no solution over the algebraic closure $\overline{\mathbb{F}}$ (i.e. the system is unsolvable). We usually shorten the polynomial equation $p = 0$ to just p. The derivation rules of the system are the following one:

$$\frac{p}{xp} \text{ (multiplication)}, \quad \frac{p \quad q}{ap + bq} \text{ (linear combination)}$$

where x is any formal variable, $a, b \in \mathbb{F}$ and p, q are polynomials over $\mathbb{F}$.

When refuting Boolean systems of equations it is common to include the Boolean axioms $x_i^2 - x_i$. Because we do *not* always include these (esp. for TI) we are explicit about our use of these, but do not assume they are built into the proof system—that is, if we are assuming them as axioms, we say so.

A *PC derivation (or proof)* of a polynomial q from a set of polynomials $\mathcal{P}$ is a sequence of polynomial equations $p_1, \ldots, p_m$ ending with the polynomial q (so p_m is q) and where each p_i, $i \in [m]$, is either an *axiom* p for $p \in \mathcal{P}$, or is obtained from previous equations in the refutation by multiplication or linear combination. We denote this by writing $\mathcal{P} \vdash q$. Observe that if p is derivable in PC and q is a polynomial then, by repeated applications of multiplication and linear combination rules, we can derive pq. We often use this generalization of the multiplication in our proofs without mention.

A *PC refutation* is just a PC proof of the polynomial 1. The *degree* of a PC derivation is the maximal degree of a polynomial used in the proof. The *size* of

a polynomial p is the number of terms in p. The *size* a PC derivation $p_1, \ldots, p_m$ is the sum of the sizes of the polynomials $p_1, \ldots, p_m$.

For our upper bound in Theorem 1.2, we also consider another algebraic proof system, known as *Nullstellensatz* (NS), to certify unsolvability of sets of polynomial equations. Nullstellensatz is defined in a static form as follows: a refutation of a list $\mathcal{P} = (p_1, \ldots, p_m)$ of polynomial equations over variables $x_1 \ldots, x_n$ is given by the list of polynomials $\mathcal{Q} = (q_1, \ldots q_m)$ such that

$$\sum_{i \in [m]} p_i q_i = 1$$

The *degree* of a NS refutation is the maximal degree of a polynomial in $\mathcal{P} \cup \mathcal{Q}$. The size of NS proof is the sum of the number of monomials appearing in the polynomials $q_1, \ldots, q_m$.

Sum-of-Squares (SOS) is a static proof system for certifying the unsolvability of systems of polynomial equations and polynomial inequalities, where polynomials are usually over the ring $\mathbb{R}[x_1 \ldots, x_n]$.

A polynomial p is a *sum-of-squares* polynomial if it is in the form $p = \sum_i r_i^2$ and the r_i's are polynomials as well. Given a system made by a set of polynomial equations $\mathcal{P} = \{p_1 = 0, \ldots p_m = 0\}$ and a set $\mathcal{Q} = \{q_1 \geq 0, \ldots q_k \geq 0\}$ of polynomial inequalities, a *sum-of-squares* proof of the polynomial inequality $p \geq 0$ from $\mathcal{P} \cup \mathcal{Q}$ is given by the formal identity

$$p = s_0 + \sum_{i \in [k]} s_i q_i + \sum_{j \in [m]} t_j p_j$$

where $s_0, s_1, \ldots, s_k$ are sum-of-squares polynomials, while $t_1, \ldots, t_m$ are arbitrary polynomials. When the system $\mathcal{P} \cup \mathcal{Q}$ is unsatisfiable, a *refutation* of $\mathcal{P} \cup \mathcal{Q}$ is a proof of the inequality $-1 \geq 0$, that is for p the constant polynomial -1. The *degree* of the proof is the $\max\{\deg(p), \deg(s_0), \deg(s_i) + \deg(q_i), \deg(t_j) + \deg(p_j) | i \in [k], j \in [m]\}$.

Definition 2.1 (PC reduction between systems of polynomials, cf. [BGIP01, Sec. 3])**.** Let $P(x_1, \ldots, x_n)$ and $Q(y_1, \ldots, y_m)$ be two sets of polynomials over a field $\mathbb{F}$. P is (d_1, d_2)-*reducible* to Q if:

1. For each $i \in [m]$ there is a polynomial $r_i(\mathbf{x})$ of degree at most d_1 (which we think of as defining y_i in terms of the $\mathbf{x}$ variables);
2. There exists a degree d_2 PC derivation of $Q(r_1(\mathbf{x}), \ldots, r_m(\mathbf{x}))$ from polynomials $P(\mathbf{x})$.

Lemma 2.2 ([BGIP01, Lem. 1])**.** *If $P(\mathbf{x})$ is (d_1, d_2)-reducible to $Q(\mathbf{y})$ and there is a degree d PC refutation of $Q(\mathbf{y})$, then there is a degree $\max(d_2, d_1 d)$ refutation of $P(\mathbf{x})$.*

In their paper, they typically only applied this to systems of equations which were known to be unsatisfiable (such as PHP and Tseitin tautologies), whereas in our paper we have several situations we want to combine the above notion together with the usual notion of many-one reduction. We encapsulate this in

the following definition. We say a decision problem Π is a *polynomial solvability problem* over a field $\mathbb{F}$ if all valid instances of the problem are systems of polynomial equations over $\mathbb{F}$, and the problem is to decide whether such a system of equations has solutions over the algebraic closure $\overline{\mathbb{F}}$. Thus, the difference between multiple polynomial solvability problems is just *which* systems of equations are valid inputs.

Definition 2.3 (PC many-one reduction)**.** Let Π_1, Π_2 be two polynomial solvability problems over a field $\mathbb{F}$. We say that Π_1 (d_1, d_2)-*many-one reduces* to Π_2 if there is a polynomial-time many-one reduction ρ from Π_1 to Π_2, such that for all unsatisfiable instances $\mathcal{F}$ of Π_1, $\mathcal{F}$ (d_1, d_2)-reduces to $\rho(\mathcal{F})$. When this occurs with $d_1, d_2 = O(1)$, we write

$$\Pi_1 \leq_m^{PC} \Pi_2.$$

2.2 Linear algebra and tensors

Given three vector spaces U, V, W over a field $\mathbb{F}$, a 3-tensor is an element of the vector space $U \otimes V \otimes W$, whose dimension is $(\dim U)(\dim V)(\dim W)$. If e_i is the i-th standard basis vector, then a basis for $U \otimes V \otimes W$ is given by the vectors $\{e_i \otimes e_j \otimes e_k\}$. One may also interpret the symbol $\otimes$ more concretely as the Kronecker product, in which $e_i \otimes e_j \otimes e_k$ represents a 3-way array whose only nonzero entry is in the (i, j, k) position. The vector space of such 3-way arrays (with coordinate-wise addition) is isomorphic to $U \otimes V \otimes W$.

The rank of a tensor $T \in U \otimes V \otimes W$ is the minimum r such that $T = \sum_{i=1}^{r} u_i \otimes v_i \otimes w_i$ for some vectors u_i, v_i, w_i.

Two $n \times m \times p$ 3-tensors $T, T' \in U \otimes V \otimes W$ are isomorphic if there exist matrices $X \in \mathrm{GL}(U), Y \in \mathrm{GL}(V), Z \in \mathrm{GL}(W)$ such that $(X, Y, Z) \cdot T = T'$, where the latter is shorthand for (1). If we treat T, T' as given non-isomorphic tensors, then we may treat (1) as a system of equations in the $n^2 + m^2 + p^2$ variables $X_{ii'}, Y_{jj'}, Z_{kk'}$. To enforce that these variable matrices are invertible, we furthermore introduce three additional sets of variables X', Y', Z' meant to be the inverse matrices, and include also the equations

$$XX' = X'X = I_n \qquad YY' = Y'Y = I_m \qquad ZZ' = Z'Z = I_p,$$

where I_n denotes the $n \times n$ identity matrix, which is Id_U in any basis. (We could have instead introduced new variables such as δ and the equation $\det(X)\delta = 1$, however, the latter equation is degree n, whereas the above equations all have degree $O(1)$, which is more desirable from the point of view of algebraic proof complexity.)

2.3 Polynomial encodings and the inversion principle

Some principles of linear algebra can be formulated as tautologies in propositional logic and therefore also as a set of polynomial equations. In this paper we preliminarily consider two such principles.

Rank Principle. As a first example we consider a set of unsatisfiable polynomials encoding the principle that the product of a $n \times r$ matrix X by a $r \times n$ matrix Y cannot be the identity matrix whenever $r < n$. We consider variables

$x_{i,k}, y_{j,k}$ for $i, j \in [n]$ and $k \in [r]$, where $r < n$ to encode X and Y. Then the polynomial encoding is:

$$\mathbb{I}(r,n) := \sum_{k\in[r]} x_{i,k}y_{j,k} - \delta_{i,j} \qquad i,j \in [n]$$

where $\delta_{i,j} = 1$ if $i = j$ and 0 otherwise. This set of polynomials is clearly unsatisfiable as long as $r < n$.

Inversion Principle. The second principle encodes the invertibility of a square $n \times n$ matrix A, expressing the tautology that $AB = I \to BA = I$ where A, B are $n \times n$ matrices and I is the identity matrix. Stephen A. Cook suggested this principle as a tautology that may be hard to prove in several proof systems.

Let $a_{i,j}, b_{i,j}$ be formal variables encoding respectively the (i,j)-th entries of A and B. We represent the fact that $AB = I$ as the set of degree 2 polynomials

$$\sum_{k\in[n]} a_{i,k}b_{k,j} - \delta_{i,j} \qquad i,j \in [n],$$

where $\delta_{i,j} = 1$ if $i = j$ and 0 otherwise. We denote this set of polynomials by $AB = I$. In Section 3, we study the degree complexity of $AB = I \vdash BA = I$, that is of PC derivations of the polynomials $BA = I$ from the polynomials $AB = I$.

In view of the results we obtain in Section 3, in Section 1.2 we considered a *polynomial rule schema* of the form

$$\frac{AB = I}{BA = I}$$

which we call the *Inversion Rule* (INV) meant to be added to PC as an extra rule. We make this slightly more precise here.

A polynomial instantiation τ of the polynomials $AB = I$ is a substitution of polynomials $p_{i,j}, q_{i,j}$ to variables $a_{i,j}$ and $b_{i,j}$. In PC+INV a polynomial p is derivable from a set of polynomials $\mathcal{P}$ if

1. p is an axiom, or $p \in \mathcal{P}$;
2. p is obtained by multiplication or linear combination from previous polynomials in the proof;
3. p is a polynomial among a polynomial instantiation τ of $BA = I$, given that among the polynomials previously derived in the proof there are all the polynomials forming the instantiation τ of $AB = I$.

Pigeonhole Principle. An important role in proving the results in Section 3 is played by the well-known *Pigeonhole principle* stating that any function f from $[n]$ to $[r]$ with $r < n$ has a collision, that is there are $i \neq i' \in [n]$ and a $j \in [r]$ such that $f(i) = f(i') = j$. PHP^n_r is the set of polynomials:

$$\sum_{k\in[r]} p_{i,k}{-}1, \text{ for } i \in [n], \qquad p_{i,k}p_{j,k}, \text{ for } i \neq j \in [n], k \in [r] \qquad p^2_{ij}{-}p_{ij}, \text{ for } i \in [n], j \in [r]$$

Razborov [Raz98] additionally included the "functional equations" (encoding that each pigeon cannot be matched to more than one hole):

$$p_{i,k}p_{i,k'}, \text{ for } i \in [n], k \neq k' \in [r].$$

3 Linear algebra warm-up: PC for matrices

Two matrices $M, M' \in U \otimes V$ are isomorphic as tensors if they are equivalent as matrices, meaning under left- and right-multiplication by invertible matrices $X \in \mathrm{GL}(U), Y \in GL(V)$, that is,

$$XMY = M'.$$

Since we want X, Y to be invertible, we also introduce variable matrices X', Y' as before, together with the equations

$$XX' = X'X = \mathrm{Id}_U \qquad YY' = Y'Y = \mathrm{Id}_V .$$

Then by left multiplying our initial matrix equation by Y', we may replace it with the new matrix equation

$$XM = M'Y'.$$

The latter has the advantage of being linear in X and Y', but the quadratic equations $XX' = \mathrm{Id}_U, YY' = \mathrm{Id}_V$ still make even this case not totally obvious.

3.1 A trick for PC degree

If our focus is on PC *degree*, we note that the degree of the equations is unchanged if we first left- or right-multiply M, M' by invertible scalar matrices. For example, if we replace M by $\overline{M} = AMB$ with $A, B \in \mathrm{GL}(U)$, then we may replace X by $\overline{X} := XA^{-1}$, Y by $\overline{Y} := B^{-1}Y$. Then we have $\overline{M} \cong M$, so $\overline{M} \cong M'$ iff $M \cong M'$. Furthermore, since the transformation $X \mapsto XA^{-1}$, $Y \mapsto B^{-1}Y$ is linear and invertible, any PC proof that $\overline{M} \ncong M'$ can be transformed by the inverse linear transformation into a PC proof that $M \ncong M'$ of the same degree.

Now, for matrices under this equivalence relation, we have a normal form, namely every matrix M is equivalent to a diagonal matrix with $\mathrm{rk}(M)$ 1s on the diagonal and all the remaining entries 0, that is, $\sum_{i=1}^{\mathrm{rk}(M)} e_i \otimes e_i = I_r \oplus 0$, where the latter 0 denotes a 0 matrix of appropriate size $(n-r) \times (m-r)$. So by using the preceding trick, we may put both M and M' in this form. The two are isomorphic iff $\mathrm{rk}(M) = \mathrm{rk}(M')$, so for PC degree we have now reduced to the case of showing that $I_r \oplus 0$ and $I_{r'} \oplus 0$ are not isomorphic when $r \neq r'$.

Note that, aside from the equations saying X and Y are invertible, this is almost identical to the Rank Principle (see Section 2.3). In the rest of this section we will prove PC lower bounds on both the Rank Principle and the Inversion Principle. Here, we show that the addition of these invertibility axioms in fact makes 2TI much easier in PC than the Rank Principle or 3TI.

Proposition 3.1. *Let M, M' be two $n \times m$ matrices of ranks r, r' respectively, with $r' > r$. Then, over any field whose characteristic does not divide $r' - r$, the following equations have a degree 3 PC refutation and a degree 4 NS refutation:*

$$XMY^T = M' \qquad XX' = X'X = \mathrm{Id}_n \qquad YY' = Y'Y = \mathrm{Id}_m \,.$$

For those familiar with the low-degree PC proof of the functional onto-PHP, the following proof is similar.

Proof idea. By the observations in Section 3.1, we may assume without loss of generality (from the point of view of PC degree) that $M = \mathrm{Id}_r \oplus 0_{n-r \times m-r}$ and $M' = \mathrm{Id}_{r'} \oplus 0_{n-r' \times m-r'}$.

Write $X = \begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{bmatrix}$ where the top-left block X_{11} has size $r' \times r$, and similarly write $Y = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$ where Y_{11} has size $r' \times r$. In this notation, the matrix equation $XMY^T = M'$ becomes the equations

$$\begin{aligned} XMY^T &= \begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{bmatrix} \begin{bmatrix} \mathrm{Id}_r & \\ & 0_{(n-r)\times(m-r)} \end{bmatrix} \begin{bmatrix} Y_{11}^T & Y_{21}^T \\ Y_{12}^T & Y_{22}^T \end{bmatrix} \\ &= \begin{bmatrix} X_{11} & 0 \\ X_{21} & 0 \end{bmatrix} \begin{bmatrix} Y_{11}^T & Y_{21}^T \\ 0 & 0 \end{bmatrix} \\ &= M' = \begin{bmatrix} \mathrm{Id}_{r'} & \\ & 0_{(n-r')\times(m-r')} \end{bmatrix} \end{aligned}$$

which becomes the four matrix equations

$$X_{11}Y_{11}^T = \mathrm{Id}_{r'} \qquad X_{11}Y_{21}^T = 0 \qquad X_{21}Y_{11}^T = 0 \qquad X_{21}Y_{21}^T = 0. \tag{2}$$

Note that so far our PC proof hasn't actually done anything—it is all just notation, and all in the same degree we started with (degree 2).

Then, using the equations $XX' = \mathrm{Id}_n$ and $YY' = \mathrm{Id}_m$, we will derive that $Y_{11}^T X_{11} = \mathrm{Id}_r$. Then we derive 1 as

$$\frac{1}{r - r'} \left(\mathrm{Tr}(X_{11}Y_{11}^T - \mathrm{Id}_{r'}) - \mathrm{Tr}(Y_{11}^T X_{11} - \mathrm{Id}_r) \right).$$

The point here is that trace is additive and cyclically invariant, so $\mathrm{Tr}(X_{11}Y_{11}^T) \equiv \mathrm{Tr}(Y_{11}^T X_{11})$, identically as polynomials, so there is no further derivation needed. □

Proof. The proof starts using the first part of the proof idea above, so we continue from Equation (2) with the notation introduced above. In the remainder of the proof, we will derive $Y_{11}^T X_{11} = \mathrm{Id}_r$. Then the last paragraph of the proof idea will complete the proof.

To derive $Y_{11}^T X_{11} = \mathrm{Id}_r$, we will use the invertibility equations (those involving X' and Y'). Write $X' = \begin{bmatrix} X'_{11} & X'_{12} \\ X'_{21} & X'_{22} \end{bmatrix}$, where X'_{11} has size $r \times r'$ (NB: the size is the "transpose" of the size of X_{11}) and similarly for Y'.

From considering the upper-left $r \times r$ block of the matrix equation $X'X = \mathrm{Id}_n$, we get

$$X'_{11}X_{11} + X'_{12}X_{21} = \mathrm{Id}_r \,.$$

Right multiplying by Y_{11}^T, we get

$$X'_{11}X_{11}Y_{11}^T + X'_{12}X_{21}Y_{11}^T = Y_{11}^T.$$

But now we can subtract from this X'_{11} times the equation $X_{11}Y_{11}^T = \mathrm{Id}_{r'}$, and also X'_{12} times the equation $X_{21}Y_{11}^T = 0$ to get

$$X'_{11} = Y_{11}^T. \tag{3}$$

Similarly, considering the upper-left $r \times r$ block of the matrix equation $Y'Y = \mathrm{Id}_m$, we get $Y'_{11}Y_{11} + Y'_{12}Y_{21} = \mathrm{Id}_r$. For consistency with the notation above, we take the transpose of this entire equation (in PC, this is essentially a null-op—we are just re-arranging how we are viewing a set of $(r')^2$ equations on the page), to get:

$$Y_{11}^T(Y'_{11})^T + Y_{21}^T(Y'_{12})^T = \mathrm{Id}_r\,.$$

Left multiplying by X_{11}, we get

$$X_{11}Y_{11}^T(Y'_{11})^T + X_{11}Y_{21}^T(Y'_{12})^T = X_{11}.$$

Now, right-multiplying the equation $X_{11}Y_{11}^T = \mathrm{Id}$ by $(Y'_{11})^T$, and right-multiplying the equation $X_{11}Y_{21}^T = 0$ by $(Y'_{12})^T$ and subtracting both of these from the above, we get

$$(Y'_{11})^T = X_{11}. \tag{4}$$

Next, we derive $M - X'M'(Y')^T = 0$, as follows: left-multiplying $XMY^T - M'$ by X', and subtract from it $(X'X - I)$ times MY^T, to get $-X'M' + MY^T$. Now right-multiply the latter by $(Y')^T$ and subtract from it M times $(Y^T(Y')^T - I)$, yielding $-X'M'(Y')^T + M$. Now multiply by -1.

Now, from $X'M'(Y')^T = M$, as at the beginning of the proof, we derive that $X'_{11}(Y'_{11})^T = \mathrm{Id}_r$. But above we have derived that $X'_{11} = Y_{11}^T$ and $(Y'_{11})^T = X_{11}$, so from the preceding three equations we get $Y_{11}^T X_{11} = \mathrm{Id}_r$, as claimed. This completes the PC proof.

Let us unroll the PC proof to derive a Nullstellensatz proof (here we underline the use of original equations):

$$r - r' = \mathrm{Tr}(\underline{X_{11}Y_{11}^T - \mathrm{Id}_{r'}}) - \mathrm{Tr}(Y_{11}^T X_{11} - \mathrm{Id}_r)$$

Now we focus on the NS derivation of $Y_{11}^T X_{11} - \mathrm{Id}_r$. Since the trace is linear, and we are focusing on degree, this is without loss of generality. We have:

$$\begin{aligned}
Y_{11}^T X_{11} - \mathrm{Id}_r &= \left(X'_{11}(Y'_{11})^T - \mathrm{Id}_r\right) - (X'_{11} - Y_{11}^T)(Y'_{11})^T - Y_{11}^T\left((Y'_{11})^T - X_{11}\right) \\
&= \Big(-X'\underline{(XMY^T - M')}Y'^T + \underline{(X'X - I)}MY^T(Y')^T + M\underline{(Y^T(Y')^T - I)}\Big)_{11} \\
&\quad + \Big(\underline{(X'_{11}X_{11} + X'_{12}X_{21} - \mathrm{Id}_r)}Y_{11}^T - X'_{11}\underline{(X_{11}Y_{11}^T - \mathrm{Id}_{r'})} - X'_{12}\underline{(X_{21}Y_{11}^T)}\Big)(Y'_{11})^T \\
&\quad - Y_{11}^T\left(X_{11}\underline{(Y_{11}^T(Y'_{11})^T + Y_{21}^T(Y'_{12})^T - \mathrm{Id}_r)} - \underline{(X_{11}Y_{11}^T - \mathrm{Id})}(Y'_{11})^T - \underline{(X_{11}Y_{21}^T)}(Y'_{12})^T\right).
\end{aligned}$$

This is visibly degree 4. □

3.2 Inversion Principle implies the Rank Principle

Lemma 3.2. *If the $r \times r$ Inversion Principle has a degree d PC derivation, then there is a degree $\max\{d, 3\}$ PC refutation of the Rank Principle stating that a rank r matrix is not equivalent (isomorphic) to a rank n matrix, for any $n > r$.*

If the Inversion Principle has a degree d NS derivation, then the Rank Principle has a degree $d + 2$ NS refutation.

Proof. Suppose the $r \times r$ Inversion Principle has a degree-d derivation. Consider the Rank Principle $XY = I_n$ where X is $n \times r$ and Y is $r \times n$, with $n > r$. Write

$$X = \begin{bmatrix} X_0 \\ X_1 \end{bmatrix} \text{ and } Y = \begin{bmatrix} Y_0 & Y_1 \end{bmatrix},$$

where X_0, Y_0 are $r \times r$. Then, examining the upper-left $r \times r$ corner of the original equations, we find $X_0 Y_0 = I_r$. As these are square matrices, by assumption in degree d we may then derive that $Y_0 X_0 = I_r$ as well.

Now, multiply both sides of $XY = I_n$ on the left by the matrix $\begin{bmatrix} Y_0 & 0 \\ 0 & I_{n-r} \end{bmatrix}$. The result is then the set of degree-3 equations

$$\begin{bmatrix} Y_0 X_0 \\ X_1 \end{bmatrix} \begin{bmatrix} Y_0 & Y_1 \end{bmatrix} = \begin{bmatrix} Y_0 & 0 \\ 0 & I_{n-r} \end{bmatrix}.$$

Considering the upper-right $r \times (n - r)$ block of these equations, we find the equations $Y_0 X_0 Y_1 = 0$.

But now, from the equation $Y_0 X_0 = I_r$, we may right-multiply by Y_1 to get $Y_0 X_0 Y_1 = Y_1$. Combining with the equation at the end of the last paragraph, we then conclude $Y_1 = 0$.

Finally, consider the lower-right $(n-r) \times (n-r)$ part of the original equation $XY = I_n$, namely, $X_1 Y_1 = I_{n-r}$. We had already derived $Y_1 = 0$, which we can then left-multiply by X_1 to get $X_1 Y_1 = 0$. Considering any diagonal entry of these two equations, we then derive the contradiction $1 = 0$.

To see the NS certificate, we unwrap the above proof. First write $Y_0 X_0 - I_r$ as a linear combination of the equations $X_0 Y_0 - I_r$ with polynomial coefficients, in total degree d. Among our starting equations in the Rank Principle, we have $X_0 Y_1$ and $X_1 Y_1 - I_{n-r}$. Then the following linear combination has degree 2 more than $Y_0 X_0 - I_r$, and derives 1 in any of its diagonal entries:

$$-X_1 Y_0 \underline{X_0 Y_1} + X_1 (Y_0 X_0 - I_r) Y_1 + \underline{(X_1 Y_1 - I_{n-r})}.$$

□

Observation 3.3. *The $n \times n$ Inversion Principle has a proof of degree $2n + 2$.*

Proof. The idea is to use Laplace expansion. We spell out the details.

We start with $XY = I_n$, where X and Y are $n \times n$ matrices of variables. Left-multiply by Y to get $YXY = Y$, and then right multiply by $Adj(Y)$ (whose entries are the $(n-1) \times (n-1)$ cofactors of Y, hence have degree $n - 1$) to get $YXY\,Adj(Y) = Y\,Adj(Y)$. Now, by Laplace expansion, we have $Y\,Adj(Y) \equiv \det(Y) I_n$, so we get $YX \det(Y) = \det(Y) I_n$.

Next, starting from $XY = I_n$ and expanding out the determinant term-by-term, we derive $\det(XY) = 1$. (Note that here, we are not simply applying

the determinant to the matrix $XY - I$, as that would give us the value of the characteristic polynomial evaluated at 1. Instead, we repeatedly use that from $a - b = 0$ and $c - d = 0$ we can derive $ac - bd = 0$ as $\underline{(a-b)}c + b\underline{(c-d)}$. Similarly, we can derive $(a + c) - (b + d) = 0$ as $\overline{(a - b) + (c - d).}$ Now, since $\det(XY) \equiv \det(X)\det(Y)$ identically as polynomials, we have derived $\det(X)\det(Y) = 1$ in degree n.

Now, from $YX\det(Y) - \det(Y)I_n$ in the first paragraph, we multiply by $\det(X)$ to get $(YX - I_n)(\det(X)\det(Y))$. From $\det(X)\det(Y) - 1$ in the second paragraph, we multiply by $-(YX - I_n)$ and add to the preceding to get $YX - I_n$, all in degree at most $2n + 2$. □

3.3 Lower bound on the Rank Principle (and Inversion Principle) via reduction from PHP

Here we show that the Rank Principle (see Section 2.3) requires large PC degree, via a reduction to the Pigeonhole Principle. For the Pigeonhole principle, a tight PC degree lower bound is known:

Theorem 3.4 (Razborov [Raz98])**.** *Any PC refutation of the Functional PHP^n_r requires degree $r/2+1$ over any field.*

We use this to show:

Theorem 3.5. *Let $n \in \mathbb{N}$, $n \geq 2$ and $1 \leq r < n$. $\mathbb{I}(r, n)$ (with or without the Boolean axioms) requires degree $r/2 + 1$ in PC over any field.*

Proof. We prove that PHP^n_r is $(1, 2)$-reducible to $\mathbb{I}(r, n)$. First we consider the following degree 1 polynomials defining x and y variables of $\mathbb{I}(r, n)$ in terms of the p variables of PHP^n_r. variables

$$x_{i,k} = y_{i,k} = p_{i,k} \qquad \text{for } i \in [n], k \in [n-1].$$

Second we show a degree 2 PC proof of $\mathbb{I}(r, n)$ from the polynomials defining the PHP^n_r. From PHP axioms $p_{i,k}p_{k,j}$ for $i, j \in [n], i \neq j$, and summing over all $k \in [r]$, we get

$$\sum_{k\in[r]} p_{i,k}p_{k,j},$$

which are exactly the axioms of $\mathbb{I}(r, n)$ for $i \neq j, i, j \in [n]$, after the substitution of variables.

For a $i \in [n]$, take the boolean axioms written in the form $p_{i,k}p_{i,k} - p_{i,k}$ and sum them over $k \in [r]$:

$$\sum_{k\in[r]} p_{i,k}p_{i,k} - \sum_{k\in[r]} p_{i,k}$$

Summing this last polynomial with the PHP axiom $\sum_{k\in[r]} p_{i,k} - 1$ we get the polynomial

$$\sum_{k\in[r]} p_{i,k}p_{i,k} - 1,$$

which is the axiom of $\mathbb{I}(r, n)$ for $i = j$ after the substitution of the variables. The proof has degree 2. The result follows immediately from Lemma 2.2 and Theorem 3.4.

□

Corollary 3.6. *Any PC proof of $AB = I \vdash BA = I$, where A, B are square $n \times n$ $\{0,1\}$ matrices requires degree $n/2 + 1$.*

Proof. Follows immediately from Theorem 3.5 and Lemma 3.2. □

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